

PREDICTING MAX TEMPERATURE FROM MIN TEMPERATURE

Analysis of Global Weather Station Data

A G E N D A

What We'll Cover

01

Dataset Overview

119,040 records from global weather stations

02

Data Cleaning

Removing bad rows, duplicates & preparing data

03

Statistical Analysis

Summary stats & distribution visualizations

04

Correlation Analysis

Measuring the MinTemp–MaxTemp relationship

05

Model Building

Linear Regression & Random Forest Regression

06

Conclusion

R^2 , MAE, RMSE & final answers

Summary of Weather Dataset

119,040

Total Records

31

Columns

2

Columns Used

Daily

Data Frequency

Research Question: Is there a relationship between daily MinTemp and MaxTemp? Can we predict MaxTemp given MinTemp?

Columns Used in Analysis

Column name	Description	Data Type
MinTemp	Minimum daily temperature recorded at the station	float64 (°C)
MaxTemp	Maximum daily temperature recorded at the station	float64 (°C)

Step - Loading & Removing Invalid Records

1

Load Dataset

Loaded 3_Summary of Weather.csv using pandas
Dataset shape: (119,040 rows × 31 columns)
Columns selected: MinTemp, MaxTemp

2

Check Missing Values

`df.isnull().sum()` — Result:
MinTemp: 0 missing ✓
MaxTemp: 0 missing ✓
Data types confirmed: both float64

3

Remove Impossible Rows

Found 32 rows where `MaxTemp < MinTemp`
(e.g. `MinTemp = 16.7°C` but `MaxTemp = -17.8°C`)
Removed all 32 bad rows → 119,008 rows remain

Duplicate Check - `df.duplicated().sum()` returned 115,428 duplicate rows
Many weather stations globally record the same temperature pair (e.g. `22.2°C / 26.7°C` is common in tropical climates)
Final Clean Dataset 3580 unique valid rows

Summary Statistics — MinTemp & MaxTemp

Statistic	MinTemp (°C)	MaxTemp (°C)
Count	3,580	3,580
Mean	5.70	17.56
Std Dev	15.97	17.99
Minimum	−38.33	−33.33
25th Percentile	−5.56	3.89
Median (50th)	7.22	20.00
75th Percentile	18.89	32.22
Maximum	34.44	50.00

11.85°C

Average daily temperature range
(MaxTemp – MinTemp)

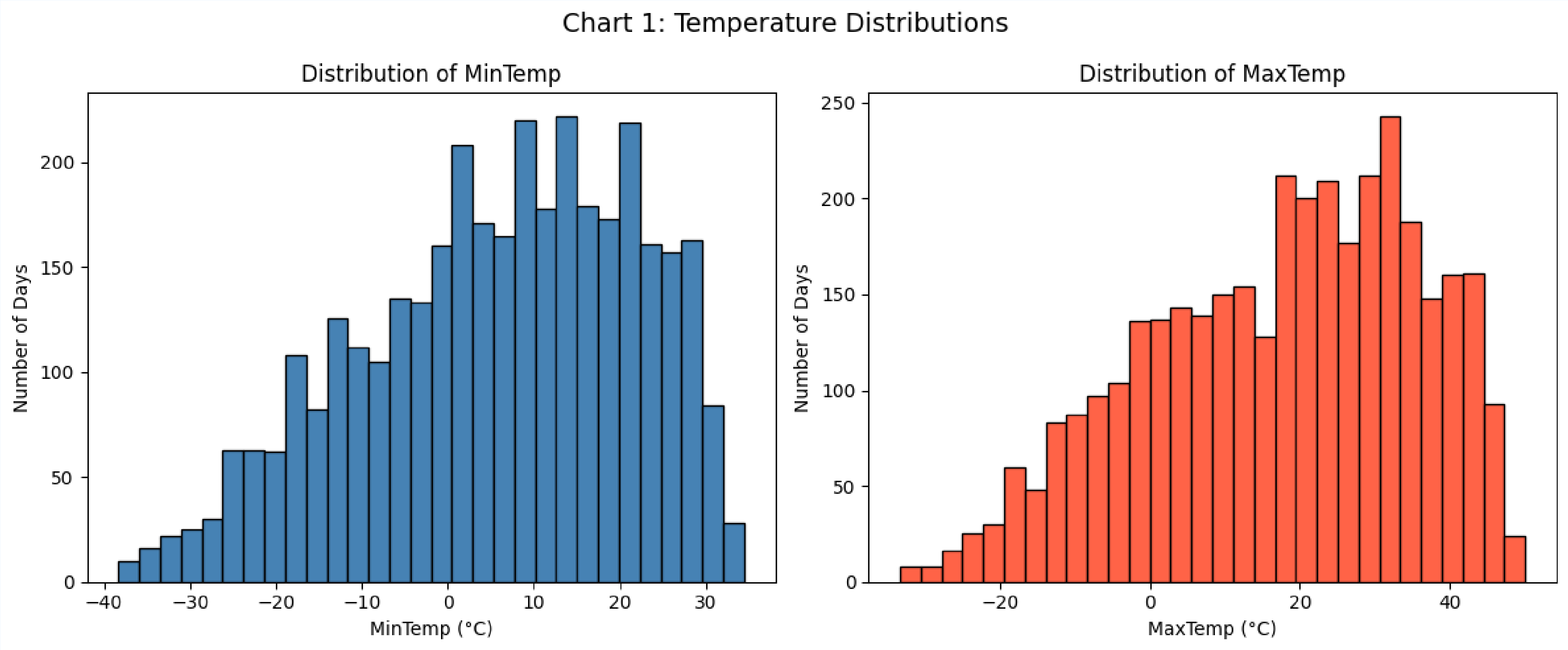
0.0°C

Minimum daily range
(some days very stable)

51.11°C

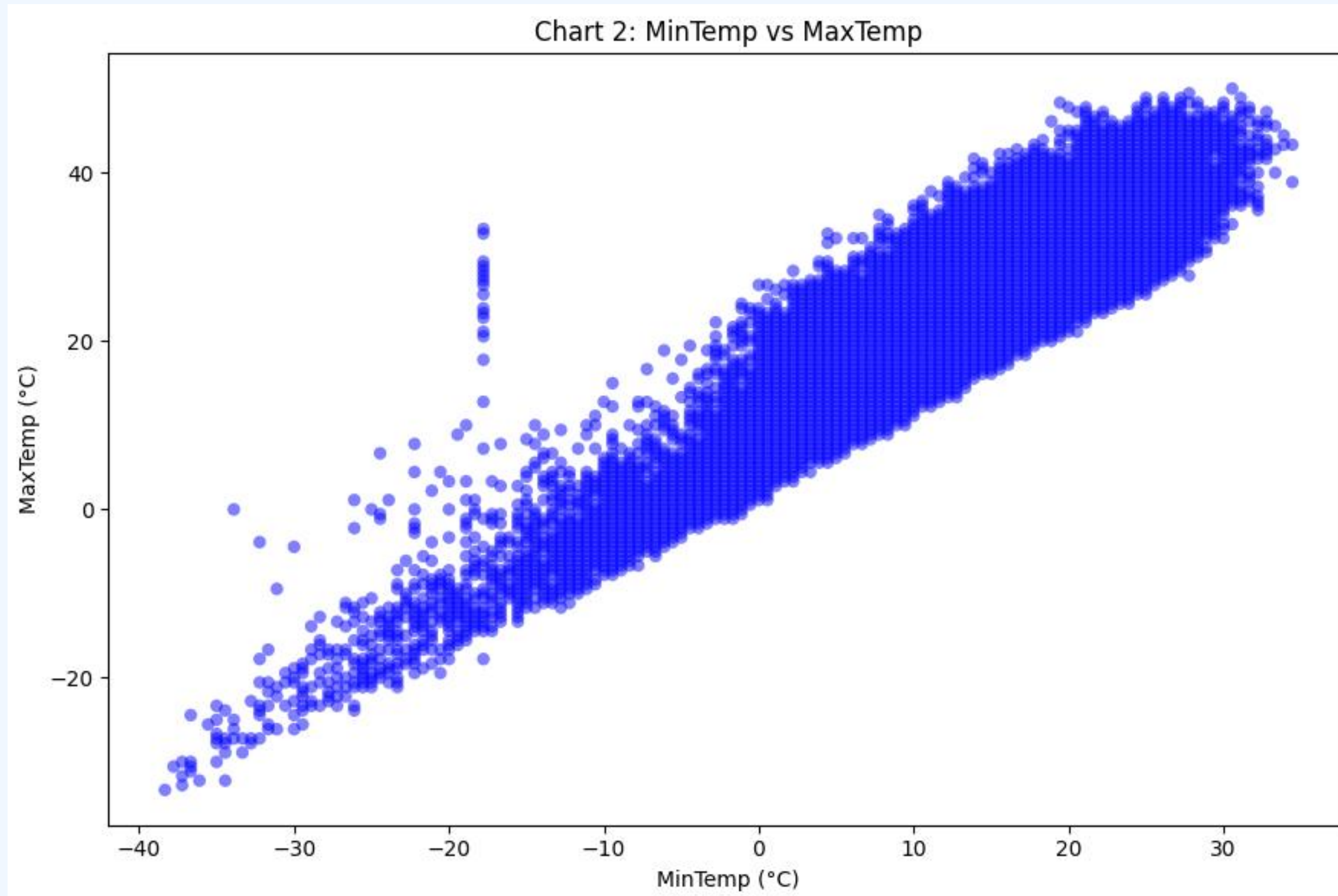
Maximum daily range
(extreme variation days)

Chart 1 — Temperature Distributions



Both distributions are approximately bell-shaped, covering a wide global range from polar cold to tropical heat.

Chart 2 — MinTemp vs MaxTemp Scatter Plot



Clear Upward Trend

As MinTemp rises, MaxTemp rises proportionally — visible linear pattern.

Global Coverage

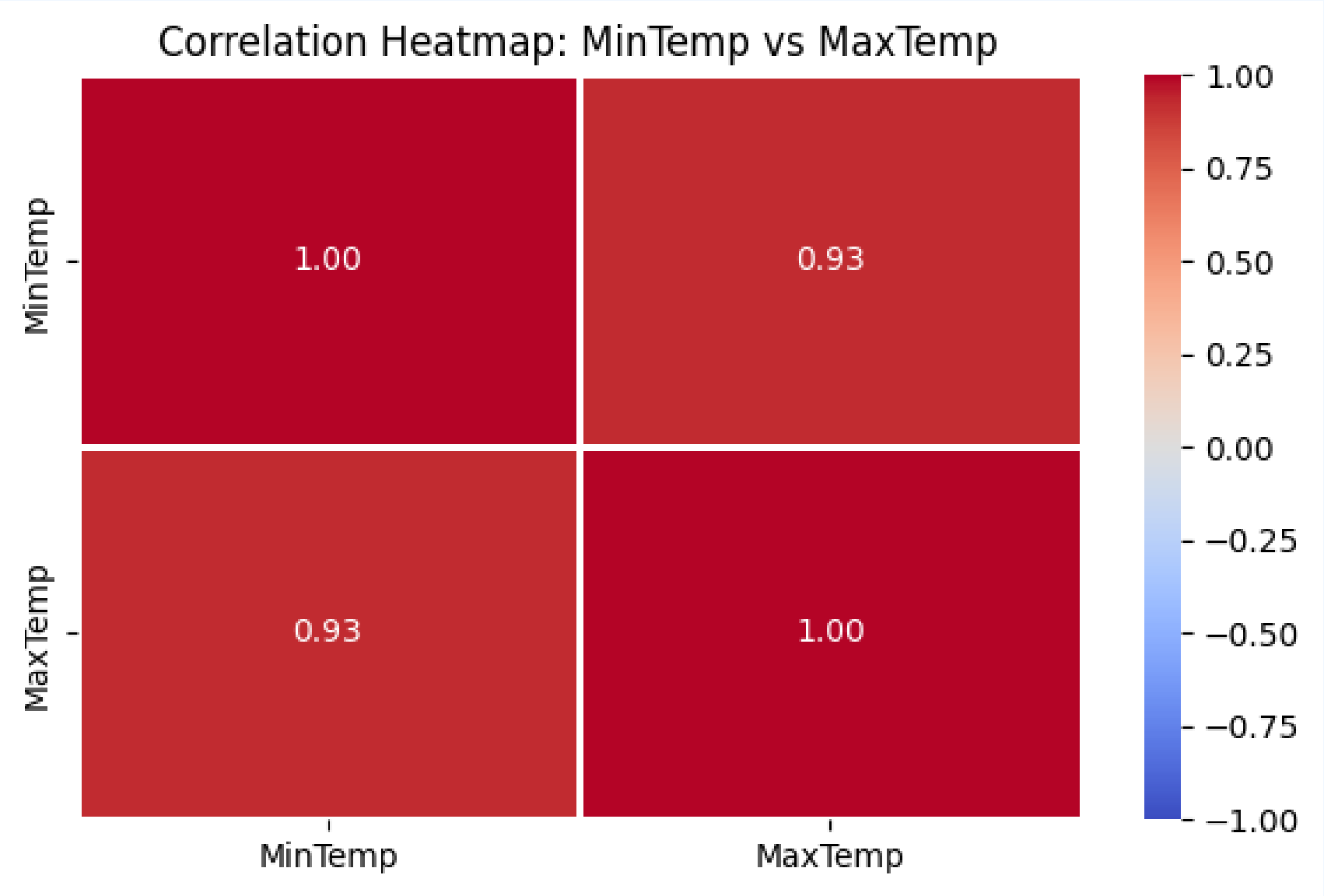
Data spans from -38°C (arctic) to $+50^{\circ}\text{C}$ (tropical) giving wide range.

Key Observation

No curved or scattered pattern — a straight line fits well.

Measuring the MinTemp–MaxTemp Relationship

Correlation Matrix



Strong Positive



Negative

ANSWER TO QUESTION 1

$$r = 0.9252$$

Pearson Correlation Coefficient

✓ YES — Very Strong Positive Relationship

When MinTemp is high, MaxTemp tends to be high too.
 $r = 0.93$ means an extremely strong linear link.

Setup — Features, Split & Algorithms

Model Configuration

Setting	Value
Input (X)	MinTemp (°C)
Output (Y)	MaxTemp (°C)
Train Split	80% → 2,864 rows
Test Split	20% → 716 rows
Random State	42 (reproducibility)

Model 1

Linear Regression

Fits a straight line through the data.
Simple, interpretable, fast.
Equation: $\text{MaxTemp} = a + b \times \text{MinTemp}$

Model 2

Random Forest

100 decision trees averaging predictions.
Handles non-linear patterns.
More complex but potentially more accurate.

Why 2 Models? → To find whether added complexity improves accuracy over a simple line.

Linear Regression — Results

Learned Equation:

MaxTemp = **11.62** + **1.04** × **MinTemp**

Example Prediction: If MinTemp = 22.22°C → Predicted MaxTemp = 11.62 + (1.04 × 22.22) = 34.82°C

Actual MaxTemp (°C)	Predicted MaxTemp (°C)
12.78	18.58
25.56	26.70
29.44	22.64
37.78	27.28
34.44	33.08

Intercept (a)

11.62 When MinTemp = 0°C, MaxTemp ≈ 11.62°C

Slope (b)

1.04 Each +1°C in MinTemp → +1.04°C in MaxTemp

Random Forest Regressor — Results

How Random Forest Works:
Builds 100 decision trees on random subsets of training data, then averages all 100 predictions. This reduces overfitting and can capture non-linear patterns that a single line misses.

Sample Prediction

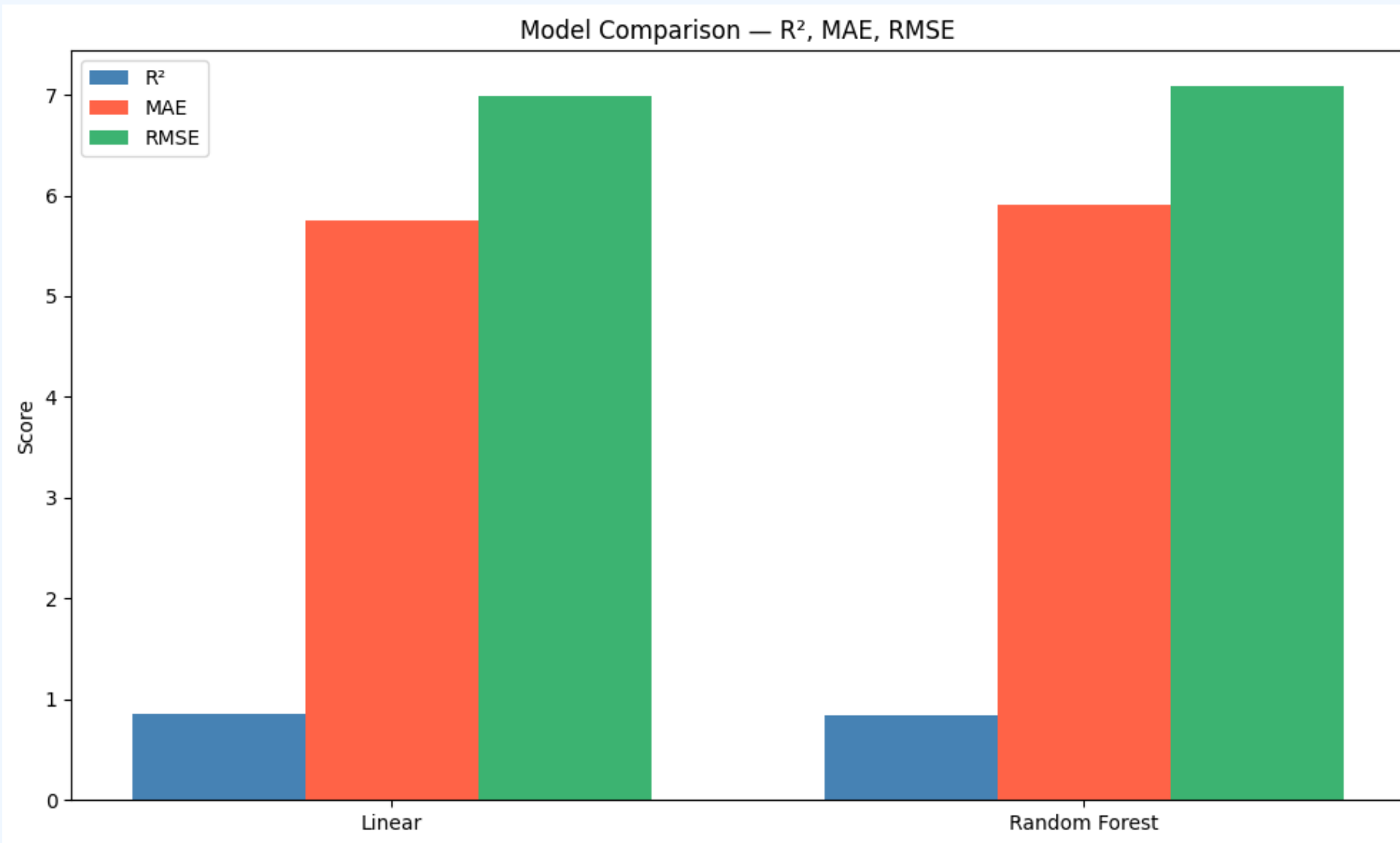
Actual MaxTemp	Predicted MaxTemp
12.777778	20.50
25.555556	29.96
29.444444	23.57
37.777778	27.09
45.555556	35.90

Compare 2 models Prediction

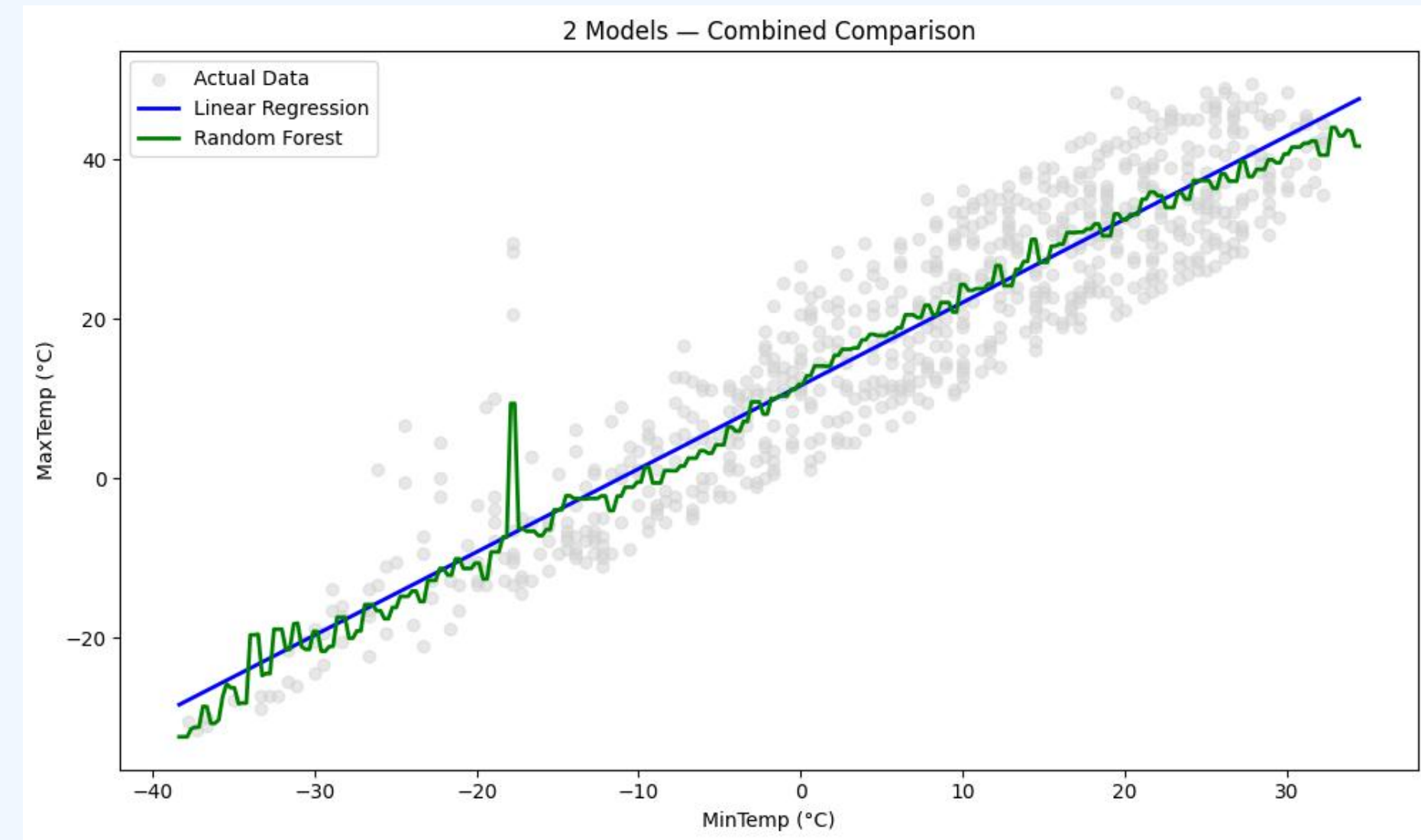
Actual (°C)	Linear Reg (°C)	Random Forest (°C)
12.78	18.58	20.50
25.56	26.70	29.96
29.44	22.64	23.57
37.78	27.28	27.09
45.56	34.24	35.90

Initial observation: The Random Forest predictions closely match Linear Regression — suggesting a strong linear relationship with minimal non-linearity.

Model Comparison — Metrics & Fit



**Model
Comparison**



**Combined Linear Regression and
Random Forest Regressor**

Linear Regression fits the data almost as well as Random Forest — with far less complexity.

Performance Metrics — Which Model Wins?

R ²	MAE (°C)	RMSE (°C)
Linear Reg 0.8499 Random Forest 0.8459 ✓ Linear wins Explains ~85% of MaxTemp variation	Linear Reg 5.7483 Random Forest 5.9089 ✓ Linear wins Average prediction error in °C	Linear Reg 6.9901 Random Forest 7.0822 ✓ Linear wins Penalises large errors more

VERDICT: Linear Regression wins on ALL 3 metrics. R² improvement of Random Forest = -0.004 (marginal). → **Linear Regression Recommended**

CONCLUSION

Answering the Research Questions

Q1 — Is there a relationship between MinTemp and MaxTemp?

✓ **YES — Very Strong Positive Relationship ($r = 0.9252$)**

When minimum temperature rises, maximum temperature rises proportionally. The correlation of 0.93 confirms an extremely strong linear link.

Q2 — Can we predict MaxTemp given MinTemp?

✓ **YES — Linear Regression: $\text{MaxTemp} = 11.62 + 1.04 \times \text{MinTemp}$**

$R^2 = 0.8499$ (explains ~85% of MaxTemp variation) · MAE = 5.75°C · RMSE = 6.99°C

Random Forest offered no meaningful improvement (R^2 diff = -0.004). Linear Regression is recommended.

Key Takeaways

- Linear Regression outperformed Random Forest on all metrics
- The simple linear model is sufficient no need for a complex model
- Future improvements: include season, location, or humidity for higher accuracy

Thank you